

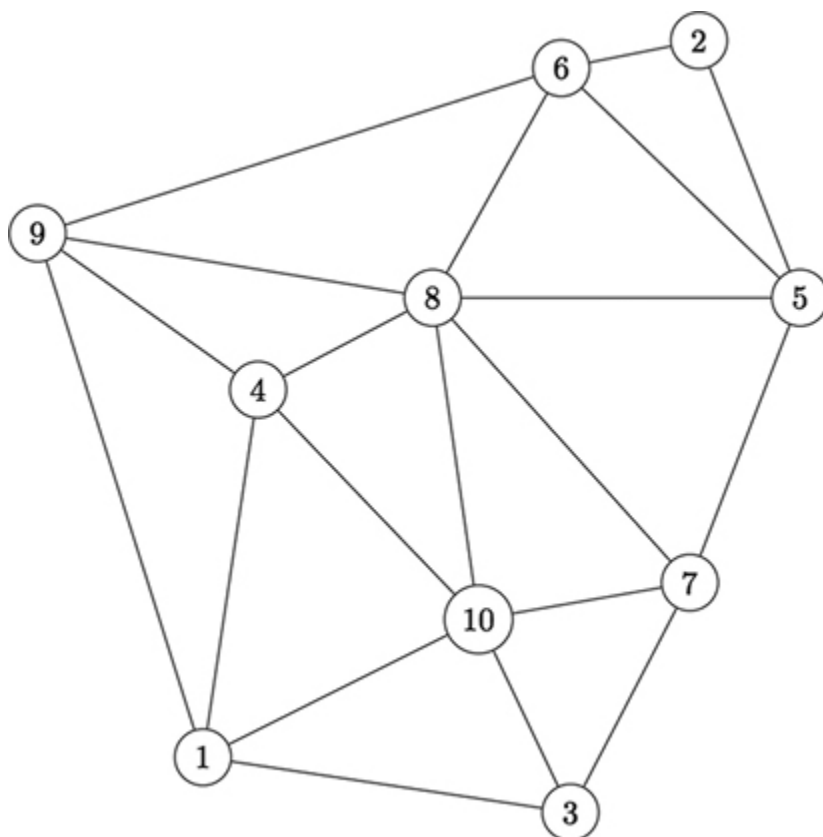


Figure 7-14: Completing the 10-point DT by reconnecting valid triangles

Voilà! We started with a 9-point DT, added a new point, and now have a 10-point DT. This process may seem straightforward. Unfortunately, as is often the case with geometric algorithms, what seems clear and intuitive to the human eye can be tricky to write code for. But let's not allow this to deter us, brave adventurers.

Implementing Delaunay Triangulations

Let's start by assuming that we already have a DT, which we'll call `delaunay`. It will be nothing more than a list of triangles. We can even start with one triangle alone:

```
delaunay = [points_to_triangle((0.2,0.8),(0.5,0.2),(0.8,0.7))]
```

Next, we'll define a point that we want to add to it, called `point_to_add`:

```
point_to_add = [0.5,0.5]
```
